

## EXPERIMENT 4

## 4 Study of the Magnetic Field Produced by an Electric Current Flowing Through a Coil of Copper Wire

### Aim

To study the magnetic field produced by an electric current flowing through a coil of copper wire.

### Requirements

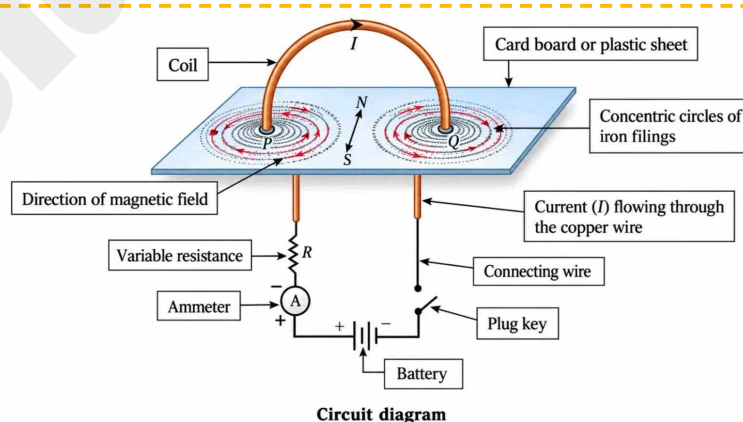
A 12 V battery, insulated copper wire (gauge 24), connecting wires, a variable resistance, a plug key, a cardboard or plastic sheet, iron filings, a magnetic needle, and a hollow tube or cylinder of cardboard.

### Procedure

1. Form a coil by winding 20 to 25 turns of insulated copper wire around a hollow cardboard tube or cylinder.
2. Pass this coil through a horizontal cardboard or plastic sheet as shown in the diagram.
3. Connect the free ends of the wire into a circuit that includes a variable resistance, an ammeter, a 12 V battery, and a plug key, as shown in the diagram.
4. Spread iron filings evenly over the cardboard or plastic sheet.
5. Close the plug key so that electric current flows through the coil.
6. Gently tap the cardboard with your finger and observe how the iron filings arrange themselves.
7. Gradually increase the current in the coil by reducing the resistance in the circuit, and observe how the arrangement of the iron filings changes.

### Diagram

(Label the various components in the given diagram - Circuit diagram)



### Observations

1. Iron filings form concentric circular paths. These concentric circles represent magnetic lines of force.
2. As we go away from the wire, the concentric circles representing the magnetic lines of force become bigger and bigger.
3. As the electric current through the coil increases, the intensity of the magnetic field also increases and the lines of force are seen more distinctly.

### Inferences / Conclusions

1. When the current flows through the coil, magnetic lines of force produced at each point around the copper wire forming the coil.
2. At a given point on the cardboard, the magnetic field produced by the current flowing through the coil is directly proportional to the current and the number of turns forming the coil.

### Precaution

Connect the various components in the electric circuit properly.

### Multiple Choice Questions

**1. Intensity of magnetic field is expressed in .....**

- (A) ampere (B) volt (C) coulomb (D) oersted

**Answer: (D) oersted**

**2. The direction of the magnetic field around a straight conductor carrying current is given by .....**

- (A) the right hand thumb rule (B) Fleming's left hand rule  
(C) Fleming's right hand rule (D) Coulomb's rule

**Answer: (A) the right hand thumb rule**

**3. .... is used to determine the presence and direction of magnetic field.**

- (A) The voltmeter (B) An ammeter (C) A galvanometer (D) The magnetic needle

**Answer: (D) The magnetic needle**

**4. In this experiment, the magnetic field at a point on the cardboard depends .....**

- (A) only on the current ( $I$ )  
(B) only on the number of turns ( $n$ ) in the coil  
(C) only on the distance ( $r$ ) of the point from the wire  
(D) on all of these, i.e.  $I$ ,  $n$  and  $r$

**Answer: (D) on all of these, i.e. l, n and r**

### **Remember**

As we go towards the centre of the wire loop, the circles representing the magnetic lines of force become so big that its arc appears as a straight line.

Date: \_\_\_\_\_

Teacher's Signature: \_\_\_\_\_